

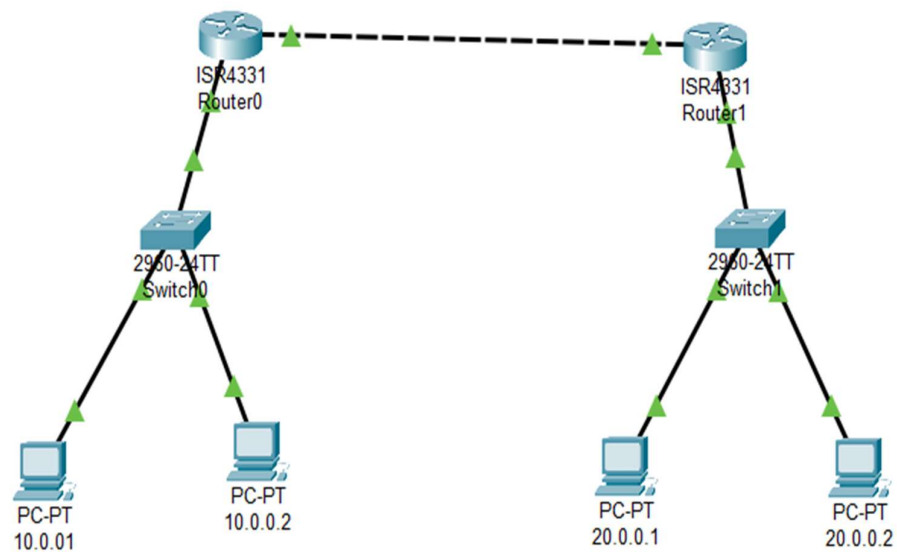
Date: 27 /08 /2026

Lab Practical #11:

Study the concept of routing using packet tracer. (Static Routing)

Practical Assignment #11:

- A. Connect the two different networks based on the calculated IP addresses and subnet using a packet tracer.





DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 27/08/2026

The screenshot shows a network topology in Packet Tracer. Two routers, ISR4331 Router0 and ISR4331 Router1, are connected via their Serial0/0/0 interfaces. Each router is connected to a 2960-XTT Switch. Each switch is connected to two PC-PTs. The PC-PTs under Router0 have IP addresses 10.0.0.1 and 10.0.0.2. The PC-PTs under Router1 have IP addresses 20.0.0.1 and 20.0.0.2. The configuration window for Router0 is open, showing the 'Config' tab. The 'Static Routes' section is visible, with the 'Network' field set to 20.0.0.0/8 via 30.0.0.2. The 'Equivalent IOS Commands' section shows the following commands:

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```

The screenshot shows the same network topology as above. The configuration window for PC-PT 10.0.0.1 is open, showing the 'Config' tab. The 'IP Configuration' section is visible, with the 'Interface' set to FastEthernet0. The 'IP Configuration' section has the following settings:

- IP Configuration: ☒ Static
- IPv4 Address: 10.0.0.1
- Subnet Mask: 255.0.0.0
- Default Gateway: 30.0.0.1
- DNS Server: 0.0.0.0

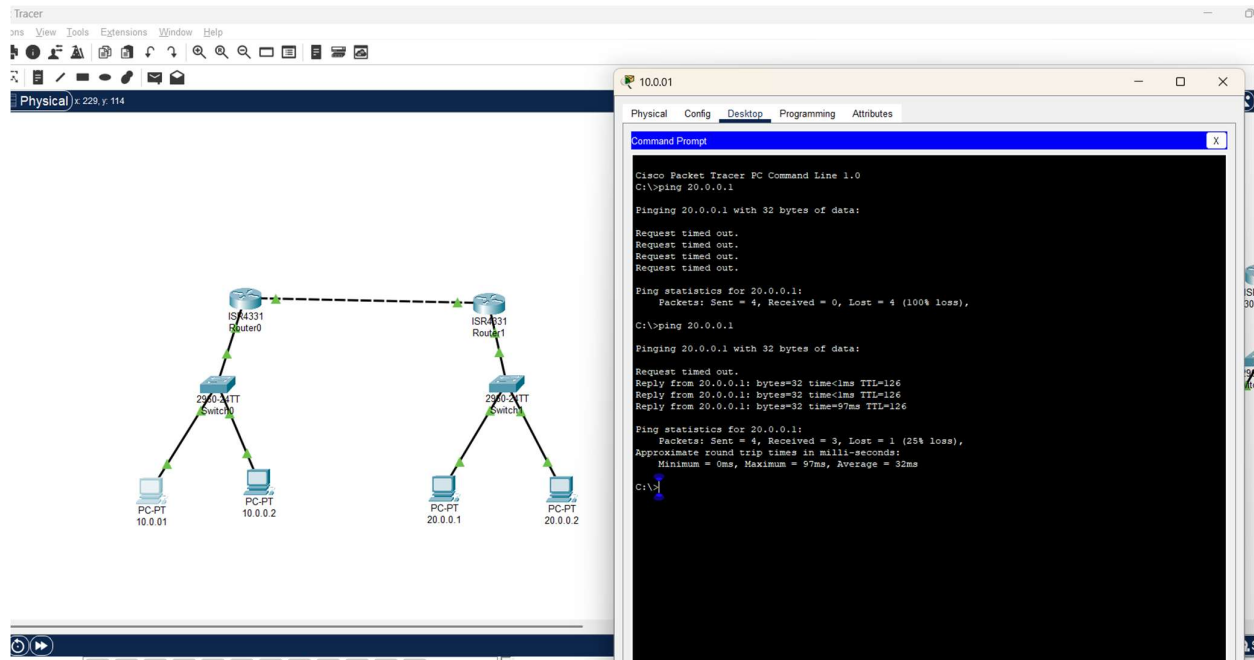
The 'IPv6 Configuration' section has the following settings:

- IPv6 Configuration: ☒ Static
- IPv6 Address: /
- Link Local Address: FE80::260:2FFF:FEC8:5559
- Default Gateway: /
- DNS Server: /

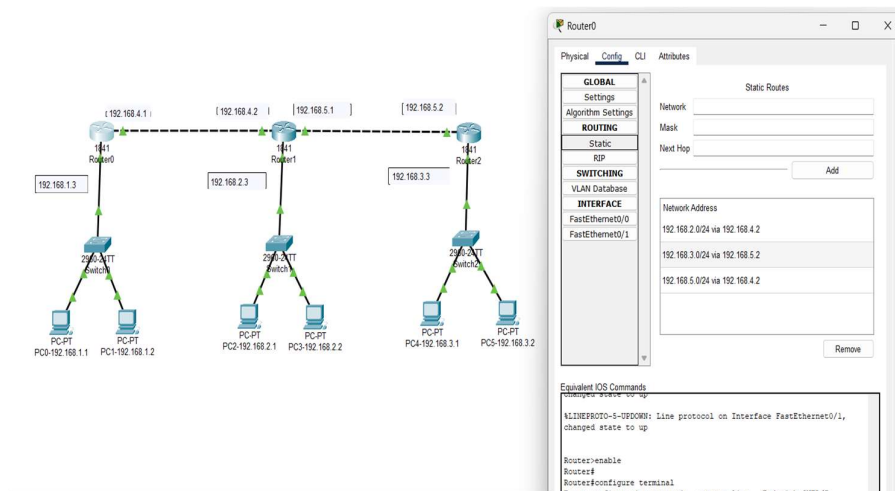
The '802.1X' section has the following settings:

- 802.1X: ☐ Use 802.1X Security
- Authentication: MD5
- Username: /
- Password: /

Date: 27/08/2026



B. Connect the three different networks based on the calculated IP addresses and subnet using a packet tracer.



Date: 27 /08 /2026

Router0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0030.F2BA.4901

IP Configuration

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

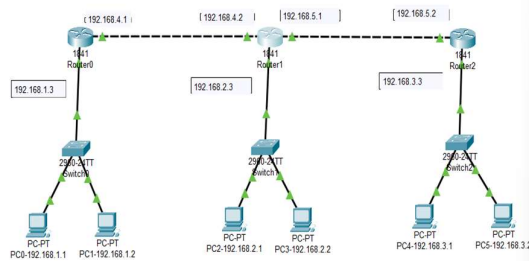
Tx Ring Limit 10

Equivalent IOS Commands

```

Router0>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
  
```

☐ Top



Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Ethernet0/1/0

Static Routes

Network:

Mask:

Next Hop:

Add

Network Address

192.168.1.0/24 via 192.168.4.1

192.168.3.0/24 via 192.168.5.2

Remove

Equivalent IOS Commands

```

Router(config)#enable
Router(config)#
Router(config)#ip
Router(config)#router rip
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
  
```

Date: 27/08/2026

